

DEVIN CHAU

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EDUCATION

San Jose State University
Bachelor of Science in Data Science
De Anza College
Associate in Computer Science

San Jose, CA
Sep. 2023 – Dec. 2025
Cupertino, CA
Aug. 2022 – Jun. 2023

EXPERIENCE

Data Science Intern

Oct. 2024 – Feb. 2025

Data Glacier

Remote

- Performed end-to-end EDA on multi-table business datasets, cleaning and joining sources to surface drivers of user behavior and revenue; delivered stakeholder-ready insights via data visualizations and presentation decks.
- Engineered a scalable ingestion and validation pipeline in Python, benchmarking Pandas vs. Dask for performance and enforcing schema validation via YAML to prevent silent data corruption.
- Automated generation of data-quality artifacts (profiling summaries, validation reports, compressed extracts) to standardize downstream reporting and improve reproducibility.

Intern

Dec. 2023 – Feb. 2024

Think Neuro

Remote

- Processed and screened 10,453 research records in R, defining inclusion criteria and constructing a structured dataset of top-cited publications for bibliometric analysis.
- Developed reproducible analysis workflows and visualizations to surface research patterns and gaps.
- Implemented validation checks and documentation standards to ensure data accuracy across the research pipeline.

PROJECTS

California Wildfire Severity Prediction | *Python, pandas, scikit-learn, XGBoost, Data Pipelines, Google Cloud*

- Performed exploratory data analysis and built an end-to-end wildfire severity prediction pipeline in Python by merging 4 public datasets into a 2,579-incident, 57-county California dataset spanning 2013–2024.
- Engineered leakage-safe weather, drought, seasonal, and demographic features using statistical methodologies, including 7-day rolling aggregates across 10 variables; reduced 41 numeric inputs to 25 final model features by removing 16 highly collinear variables.
- Developed and evaluated data models with Random Forest, XGBoost, and stacking ensemble techniques using a time-based 2013–2021 train / 2022–2024 test split, improving from a mean-baseline test R-squared of -0.16 to an XGBoost test R-squared of 0.97.

Weather and Energy Demand Pipeline | *AWS, PySpark, Python, SQL, Parquet*

- Built an AWS-based pipeline observability framework to process and analyze data quality metrics using DynamoDB logs, CloudWatch metrics, and SNS alerts, detecting issues such as duplicate keys, incomplete hourly coverage, and invalid weather readings.
- Implemented clean, reusable code for monitoring and alerting workflows, integrating DynamoDB logging and SNS notifications to ensure rapid response to pipeline failures.

Starbucks Marketing Funnel Analysis | *Python, pandas, NumPy, Matplotlib/Seaborn*

- Reconstructed per-user offer life-cycles by linking portfolio/profile/transcript data and computing funnel KPIs.
- Implemented offer attribution logic to associate events within active offer windows, handling edge cases for reliable conversion measurement.
- Produced segment-level performance cuts by offer type/channel and customer attributes to identify drop-off drivers and recommend a higher-impact promotion mix.

TECHNICAL SKILLS

Analysis: SQL (MySQL/PostgreSQL), Excel (PivotTables, XLOOKUP), Python (pandas, NumPy), R, Machine Learning, Statistics, Data Analysis, Data Modeling

Data Quality: Schema validation (YAML), ingestion benchmarking (Pandas vs. Dask), reproducible notebooks

Reporting: Matplotlib, Dash, written summaries and slide decks, Data Visualization, Tableau, Power BI

Tools: Git/GitHub, REST APIs, AWS, GCP, Azure